

CARBON FAMILY

SECTION A — NCERT STATEMENT (Q1–Q20)

Q1. How many of the following statements about occurrence, abundance and electronic configuration of Group 14 elements are correct?

1. Group 14 consists of C, Si, Ge, Sn and Pb.
2. The valence-shell configuration of Group 14 elements is ns^2np^2 .
3. Carbon occurs in both free and combined states.
4. Natural carbon contains stable ^{12}C and ^{13}C , while ^{14}C is radioactive.
5. The half-life of ^{14}C is about 5770 years.
6. Silicon is the second most abundant element in Earth's crust by mass and occurs chiefly as silica and silicates.
7. Germanium occurs only in traces.
8. Tin occurs mainly as cassiterite, SnO_2 .
9. Lead occurs mainly as galena, PbS .
10. Ultrapure Si and Ge are used in transistors and semiconductor devices.
11. Every Group 14 element occurs abundantly in the free state.
12. Carbon and silicon are metals while germanium is a non-metal.

Q2. How many of the following statements about covalent radius, ionisation enthalpy and electronegativity trends in Group 14 are correct?

1. There is a large increase in covalent radius from C to Si.
2. From Si to Pb the increase in radius is comparatively small because filled d/f shells affect shielding.
3. First ionisation enthalpy generally decreases down the group.
4. The first ionisation enthalpy falls slightly from Si to Ge to Sn.
5. A small increase from Sn to Pb is linked to poor shielding by intervening d and f electrons.
6. The first ionisation enthalpy decreases perfectly smoothly $\text{C} > \text{Si} > \text{Ge} > \text{Sn} > \text{Pb}$.
7. Group 14 elements are slightly more electronegative than corresponding Group 13 elements.
8. Electronegativity values of Si, Ge and Sn are nearly the same on the Pauling scale.
9. Pb is slightly more electronegative than Si.
10. The irregular trends among heavier members reflect differences in their inner electronic cores.
11. Carbon has the highest electronegativity value among Group 14 elements.
12. The electronegativity values of all Group 14 elements are identical.

Q3. How many of the following statements about the numerical periodic data of Group 14 elements are correct?

1. The covalent radii for C, Si, Ge, Sn and Pb are 77, 118, 122, 140 and 146 pm, respectively.
2. The first ionisation enthalpies are approximately 1086, 786, 761, 708 and 715 kJ mol^{-1} .
3. Using the values, the IE_1 order is $\text{C} > \text{Si} > \text{Ge} > \text{Pb} > \text{Sn}$.
4. The density of diamond is 3.51 g cm^{-3} , whereas that of graphite is about 2.22 g cm^{-3} .
5. The melting point of Sn is lower than that of Pb.
6. The boiling point decreases from Si to Ge to Sn to Pb.
7. The covalent radius of Ge is smaller than that of Si.
8. The first ionisation enthalpy of Pb is slightly greater than that of Sn.
9. All Group 14 elements have identical electronegativity.
10. Electrical resistivity data distinguish non-metallic/semiconducting members from metallic Sn and Pb.
11. The covalent radius of Pb is larger than that of Sn.
12. The first ionisation enthalpy of carbon is the highest among the listed Group 14 elements.

Q4. How many of the following statements about oxidation states and the inert-pair effect in Group 14 are correct?

1. The common oxidation states of Group 14 are +4 and +2.
2. Carbon can also exhibit negative oxidation states.
3. Compounds in the +4 state are generally covalent because the sum of four ionisation enthalpies is very high.
4. The tendency to show the +2 state increases in the order $\text{Ge} < \text{Sn} < \text{Pb}$.
5. The increased stability of +2 in heavier members is associated with reduced participation of ns^2 electrons in bonding.
6. Carbon and silicon mainly show +4.
7. Germanium has stable +4 compounds and only a few +2 compounds.
8. Sn(II) compounds can act as reducing agents.
9. Pb(IV) compounds can act as strong oxidising agents.
10. Pb(IV) is more stable than Pb(II) in ordinary chemistry.
11. The inert-pair effect becomes less important on descending the group.
12. The +2 oxidation state becomes progressively more important for the heavier members of Group 14.

Q5. How many of the following statements about Group 14 oxides and reactions with water and halogens are correct?

1. Members form monoxides MO and dioxides MO_2 on heating in oxygen.
2. SiO is stated to exist only at high temperature.
3. CO_2 , SiO_2 and GeO_2 are acidic dioxides.
4. SnO_2 and PbO_2 are amphoteric.
5. CO is a neutral oxide.

6. SnO and PbO are amphoteric.

7. C, Si and Ge are not affected by water under the stated conditions.
8. Tin decomposes steam to form SnO_2 and H_2 .
9. Lead is largely unaffected by water because of protective oxide film formation.
10. Every Group 14 tetrachloride is ionic.
11. The stability of dihalides increases down the group.
12. PbI_4 is a stable ordinary lead(IV) halide.

Q6. How many of the following statements about hydrolysis, complex formation and the anomalous behaviour of carbon are correct?

1. CCl_4 resists hydrolysis whereas SiCl_4 hydrolyses readily.
2. SiCl_4 hydrolysis can be rationalised through acceptance of a water lone pair and formation of higher-coordinate silicon.
3. Species such as $[\text{SiF}_6]^{2-}$, $[\text{GeCl}_6]^{2-}$ and $[\text{Sn}(\text{OH})_6]^{2-}$ illustrate higher coordination in heavier members.
4. Carbon cannot exceed covalence four in the elementary textbook model.
5. Carbon's anomaly is linked to small size, high electronegativity, high ionisation enthalpy and lack of d orbitals.
6. Carbon forms strong $\text{p}\pi\text{--p}\pi$ multiple bonds with itself and small electronegative atoms.
7. Heavier Group 14 atoms form $\text{p}\pi\text{--p}\pi$ multiple bonds more effectively than carbon.

8. $[\text{SiCl}_6]^{2-}$ is as readily formed as $[\text{SiF}_6]^{2-}$ because Cl^- is smaller than F^- .

9. The absence of low-lying valence d orbitals is part of the textbook explanation for carbon's maximum covalence of four.
10. CCl_4 does not hydrolyse readily under ordinary conditions in the textbook treatment.
11. Heavier Group 14 members can show coordination numbers greater than four.
12. $[\text{SiF}_6]^{2-}$ is cited as an example of a higher-coordinate silicon species.

Q7. How many of the following statements about catenation and multiple bonding in Group 14 are correct?

1. Catenation is the tendency of an element to form chains or rings by element–element covalent bonds.
2. Strong C–C bonding is a major reason for exceptional catenation of carbon.
3. The stated order is $\text{C} \gg \text{Si} > \text{Ge} \approx \text{Sn}$, while Pb essentially does not show catenation.
4. The E–E bond enthalpies are C–C 348, Si–Si 297, Ge–Ge 260 and Sn–Sn 240 kJ mol^{-1} .
5. Catenation increases down Group 14 as atomic size increases.
6. $\text{C}=\text{C}$, $\text{C}\equiv\text{C}$, $\text{C}=\text{O}$, $\text{C}=\text{S}$ and $\text{C}\equiv\text{N}$ are examples illustrating carbon multiple bonding.
7. The ability of carbon to catenate and form $\text{p}\pi\text{--p}\pi$ bonds contributes to allotropy.

8. Pb shows catenation more strongly than Sn.

9. A larger, more diffuse valence shell generally weakens effective $\text{p}\pi\text{--p}\pi$ overlap.

10. The C–C bond enthalpy is greater than the Si–Si bond enthalpy.

11. The Sn–Sn bond enthalpy is lower than the Ge–Ge bond enthalpy.

12. Carbon shows much greater catenation than silicon.

Q8. How many of the following statements about the structure and properties of diamond are correct?

1. Every carbon atom in diamond is sp^3 -hybridised.
 2. Each carbon is tetrahedrally bonded to four other carbon atoms.
- 3. The C–C bond length is 154 pm.**
4. Diamond is a rigid three-dimensional covalent network.
 5. Its hardness is attributed to extended directional covalent bonding.
 6. Diamond is an electrical conductor because it has delocalised π electrons.
 7. Diamond has a high melting point because many strong C–C bonds must be disrupted.
 8. Diamond is used as an abrasive and for sharpening hard tools.
 9. Diamond is structurally a layered solid held by van der Waals forces.
 10. Diamond is used as a precious stone and is measured in carats; 1 carat = 200 mg.
 11. Diamond contains directional covalent bonds throughout its lattice.
 12. The three-dimensional network is responsible for its exceptional hardness.

Q9. How many of the following statements about the structure, bonding and uses of graphite are correct?

1. Graphite has a layered structure.
2. The distance between adjacent layers is about 340 pm.
3. Each layer contains planar hexagonal rings.
4. The in-plane C–C distance is 141.5 pm.
5. Each carbon is sp^2 -hybridised and makes three σ bonds.
6. The remaining electron participates in a delocalised π system.
7. Graphite conducts electricity along the layers.
8. Weak interlayer forces make graphite soft and slippery.
9. Graphite is used as a high-temperature dry lubricant.
10. The layers are joined to one another mainly by strong covalent C–C bonds.
11. Graphite is thermodynamically the most stable allotrope under standard conditions.
12. The mobile delocalised electrons are responsible for electrical conduction along graphite sheets.

Q10. How many of the following statements about the preparation and structure of fullerenes, especially C_{60} are correct?

1. Fullerenes can be prepared by heating graphite in an electric arc in inert gas such as He or Ar.
2. The soot contains mainly C_{60} with some C_{70} and higher even-carbon fullerenes.
3. Fullerenes are pure forms of carbon with smooth structures without dangling bonds.
4. C_{60} is a cage molecule shaped like a soccer ball.
5. C_{60} contains 20 six-membered rings and 12 five-membered rings.
6. A five-membered ring in C_{60} is fused only to six-membered rings.
7. All carbon atoms in C_{60} are described as sp^2 -hybridised.
8. C_{60} has 60 vertices, each occupied by one carbon atom.
9. The two C–C distances are about 143.5 and 138.3 pm.
10. C_{60} contains only six-membered rings.
11. The $\Delta_f H^\circ$ values relative to graphite are about 1.90 kJ mol^{-1} for diamond and 38.1 kJ mol^{-1} for C_{60} .
12. Fullerenes are cage-like molecules.
13. C_{60} contains both single and double C–C bonds.

Q11. How many of the following statements about the uses and impure forms of carbon are correct?

1. Carbon black is obtained by burning hydrocarbons in a limited supply of air.
2. Charcoal and coke are produced by heating wood or coal, respectively, in the absence of air.
3. Graphite fibres in plastics give strong, lightweight composites.
4. Graphite is used for electrodes in batteries and industrial electrolysis.
5. Graphite crucibles are resistant to dilute acids and alkalis.
6. Activated charcoal adsorbs poisonous gases and is used in water purification/filters.
7. Carbon black is used as black pigment and as filler in automobile tyres.
8. Coke is used as a fuel and as a reducing agent in metallurgy.
9. Diamond is measured in carats, with 1 carat equal to 20 mg.
10. Activated charcoal is used because it is highly porous.
11. Graphite fibres embedded in plastic are used to make high-strength lightweight composites.
12. Activated charcoal is used because of its high porosity and adsorption capacity.

Q12. How many of the following statements about the preparation of carbon monoxide and industrial fuel-gas mixtures are correct?

1. Limited oxidation of carbon gives CO.

2. Pure CO can be prepared by dehydrating formic acid with concentrated H_2SO_4 at 373 K.

3. Passing steam over hot coke gives $CO + H_2$, called water gas or synthesis gas.

4. Using air instead of steam gives CO mixed mainly with N_2 , called producer gas.
5. Water gas and producer gas are important industrial fuels.
6. CO burns further to CO_2 with release of heat.
7. Water gas is a mixture of CO_2 and H_2 .
8. Producer gas is chiefly $CO + N_2$ in this description.
9. Complete combustion of carbon in excess O_2 is the preferred laboratory preparation of pure CO.
10. Producer gas is obtained when air is used instead of steam over hot carbon.
11. Water gas contains hydrogen whereas producer gas contains nitrogen as a major accompanying gas.
12. Carbon monoxide in these fuel mixtures can burn further to carbon dioxide with liberation of heat.

Q13. How many of the following statements about the structure, reducing action and toxicity of carbon monoxide are correct?

1. CO is colourless, odourless and almost insoluble in water.
2. CO is a powerful reducing agent for many metal oxides.
3. CO reduces Fe_2O_3 and ZnO under suitable heating conditions.
4. The CO molecule is represented with one σ and two π bond components.
5. A lone pair on carbon allows CO to act as a donor ligand in metal carbonyls.
6. The high toxicity of CO is associated with strong binding to haemoglobin.
7. The CO–haemoglobin complex is about 300 times more stable than the oxygen–haemoglobin complex.
8. CO is an acidic oxide that forms carbonic acid directly with water.
9. CO prevents haemoglobin from carrying oxygen efficiently.
10. CO_2 , rather than CO, is the classic gas that forms carboxyhaemoglobin.
11. Carbon monoxide acts as a donor toward certain metals because of the lone pair on carbon.
12. Carbon monoxide is used as a reducing agent in extraction of some metals from their oxides.

Q14. How many of the following statements about the preparation and acid-base chemistry of carbon dioxide and the blood buffer system are correct?

1. CO_2 forms on complete combustion of carbon and carbon-containing fuels in excess air.
2. In the laboratory, dilute HCl on $CaCO_3$ gives CO_2 .
3. Commercial CO_2 can be obtained by heating limestone.
4. CO_2 is colourless and odourless.
5. Dissolved CO_2 forms weak dibasic carbonic acid, H_2CO_3 .
6. H_2CO_3 dissociates stepwise through HCO_3^- and then CO_3^{2-} .
7. The H_2CO_3/HCO_3^- system helps buffer blood pH.

- The stated blood pH range is roughly 7.26–7.42.
- CO₂ is a neutral oxide.
- CO₂ reacts with alkalies to form carbonates/bicarbonates depending on conditions.
- Carbonic acid is weak and dibasic.
- The H₂CO₃/HCO₃[−] buffer system helps maintain the pH of blood.

Q15. How many of the following statements about the structure, environmental chemistry and uses of carbon dioxide are correct?

- Atmospheric CO₂ is removed by photosynthesis.
- Rising CO₂ from fossil-fuel combustion and limestone decomposition contributes to greenhouse effect.
- Solid CO₂ is called dry ice.
- Dry ice is used as a refrigerant for frozen foods.
- CO₂ is used to carbonate soft drinks.
- Its density and inability to support ordinary combustion allow fire-extinguisher use.
- CO₂ is used in manufacture of urea.
- Carbon in CO₂ is sp-hybridised.
- CO₂ is linear with equal C–O bonds near 115 pm and has zero dipole moment.
- CO₂ has a bent molecular shape because carbon has a lone pair.
- The elementary bonding description assigns two σ bonds and two π–π components to CO₂.
- Dry ice is obtained by rapid expansion of liquefied carbon dioxide.

Q16. How many of the following statements about the forms, structure, reactions and uses of silica are correct?

- Quartz, cristobalite and tridymite are crystalline forms of silica.
- These crystalline forms are interconvertible at suitable temperatures.
- SiO₂ is a covalent three-dimensional network.
- Each Si is tetrahedrally bonded to four oxygen atoms.
- Each oxygen bridges two silicon atoms in the network.
- The network can be viewed as containing eight-membered Si–O rings.
- Normal silica is comparatively unreactive because the Si–O bond enthalpy is high.
- Silica is attacked by HF and NaOH.
- Quartz is used as a piezoelectric material.
- Silica gel is used as a drying agent and as support for chromatography/catalysts.
- Kieselguhr is an amorphous silica used in filtration.
- SiO₂ dissolves readily in most mineral acids.

Q17. How many of the following statements about the preparation and structural architecture of silicones are correct?

- Silicones are organosilicon polymers containing –Si–O–Si– backbones.
- Simple silicone repeating units are commonly represented as (R₂SiO).
- The starting chlorosilanes have general formula R_nSiCl_{4–n}.
- Methyl chloride reacts with silicon in the presence of Cu catalyst near 573 K to give a mixture of methylchlorosilanes.
- Hydrolysis of (CH₃)₂SiCl₂ gives a silanediol-type intermediate.
- Condensation of the difunctional silanediol gives mainly chain/cyclic silicone structures.
- (CH₃)₃SiCl can terminate chains by blocking ends.
- RSiCl₃-derived units can introduce cross-linking in silicone networks.
- R₄Si is the preferred trifunctional cross-linking monomer.
- The Si–O–Si backbone is absent from silicones.
- Hydrolysis of dimethyldichlorosilane followed by condensation polymerisation yields straight-chain polymers.
- Trimethylchlorosilane is used to control chain length by blocking chain ends.

Q18. How many of the following statements about the properties and applications of silicones are correct?

- Non-polar organic groups make silicones water-repelling.
- Silicones generally have high thermal stability.
- They generally possess high dielectric strength.
- They show resistance to oxidation and many chemicals.
- They are used as sealants.
- They are used as greases/lubricants.
- They are used as electrical insulators.
- They are used for waterproofing fabrics.
- Silicones are biocompatible and are used in surgical/cosmetic applications.
- Silicones are strongly hydrophilic because every Si atom carries –OH groups in the final polymer.
- Silicones are characterised by low dielectric strength.
- Silicones are water-repelling because non-polar alkyl groups surround the Si–O backbone.

Q19. How many of the following statements about the structural unit and polymerisation of silicates are correct?

- The basic structural unit of silicates is [SiO₄]^{4−}.
- Si is tetrahedrally surrounded by four oxygen atoms.
- Silicate tetrahedra may occur as discrete units or share corners.
- Corner sharing may involve 1, 2, 3 or 4 oxygen atoms per tetrahedron.

5. Corner sharing can generate chains, rings, sheets or three-dimensional networks.
6. Negative charge is balanced by positive metal ions.
7. Sharing all four corners generates a three-dimensional network.
8. Feldspar, zeolites, mica and asbestos are examples of silicate/aluminosilicate minerals mentioned.
9. Glass and cement are cited as important man-made silicates.
10. Silicate tetrahedra polymerise mainly by sharing edges in the stated model.
11. Every silicate must contain isolated $[\text{SiO}_4]^{4-}$ ions only.
12. Sharing all four corners of the SiO_4 tetrahedra gives a three-dimensional network.

Q20. How many of the following statements about zeolites and aluminosilicates are correct?

1. Replacement of some Si by Al in a three-dimensional silica network creates an aluminosilicate framework with net negative charge.
2. Na^+ , K^+ or Ca^{2+} can balance the framework charge.
3. Feldspar and zeolites are examples of aluminosilicate frameworks.
4. Zeolites are used as catalysts in petrochemical cracking/isomerisation.
5. ZSM-5 is cited for converting alcohols directly to gasoline.
6. Hydrated zeolites can act as ion exchangers in water softening.
7. Al-for-Si substitution makes the framework positively charged.
8. Zeolites soften hard water by anion exchange only.
9. The framework charge can be balanced by mobile cations.
10. Zeolite frameworks are aluminosilicates in which cations balance the negative framework charge.
11. Zeolites are used for cracking and isomerisation of hydrocarbons.
12. Hydrated zeolites can exchange cations during hard-water softening.

SECTION B — MULTIPLE / STATEMENT-COMBINATION MCQ (Q21–Q54)

One or more of I–V may be correct; choose the option that lists exactly the correct set.

Q21. Which option lists exactly the correct statements about the occurrence, abundance and periodic data of Group 14 elements?

- I. Every member of Group 14 has the valence-shell configuration ns^2np^2 .
- II. Carbon is the second most abundant element in the Earth's crust by mass.
- III. Cassiterite is SnO_2 , galena is PbS , and germanium occurs only in traces.
- IV. The covalent radii of C, Si, Ge, Sn and Pb are about 77, 118, 122, 140 and 146 pm.

V. CF_2Cl_2 (Freon-12) has been used as a refrigerant.

(A) I, II, III and IV only **(B)** II, III and V only **(C)** I, III, IV and V only **(D)** all of I–V

Q22. Which option lists exactly the correct statements about covalent radius and ionisation enthalpy trends in Group 14?

- I. There is a large increase in covalent radius from C to Si, then only a small increase from Si to Pb.
- II. The small increase from Si to Pb is due to poor screening by filled d and f subshells.
- III. $\Delta_i H_1$ passes through a minimum near Sn and then rises slightly for Pb.
- IV. Each Group 14 element has a higher $\Delta_i H_1$ than the corresponding Group 13 element.

V. $\Delta_i H_1$ falls smoothly and regularly along $\text{C} > \text{Si} > \text{Ge} > \text{Sn} > \text{Pb}$.

(A) I, II, III and IV only **(B)** II, III and V only
(C) I, III, IV and V only **(D)** all of I–V

Q23. Which option lists exactly the correct statements about carbonate and carbon-dioxide reactions?

- I. $\text{CaCO}_3 \xrightarrow{\Delta} \text{CaO} + \text{CO}_2$ and $\text{CaCO}_3 + 2\text{HCl} \rightarrow \text{CaCl}_2 + \text{CO}_2 + \text{H}_2\text{O}$.
- II. Excess CO_2 dissolves the CaCO_3 precipitate of limewater as soluble $\text{Ca}(\text{HCO}_3)_2$.
- III. $\text{CaCO}_3 + \text{CO}_2 + \text{H}_2\text{O} \rightarrow \text{Ca}(\text{HCO}_3)_2$.
- IV. CO_2 and NaOH in 1 : 1 mole ratio give NaHCO_3 ; in 1 : 2 ratio they give Na_2CO_3 .
- V. Passing excess CO_2 into limewater makes the CaCO_3 precipitate permanent.

(A) II, III and V only **(B)** I, II, III and IV only
(C) I, IV and V only **(D)** all of I–V

Q24. Which option lists exactly the correct statements about oxidation states and the inert-pair effect in Group 14?

- I. The common oxidation states of Group 14 are +4 and +2.
- II. In ordinary lead chemistry Pb(IV) is more stable than Pb(II) .
- III. The tendency to show +2 rises $\text{Ge} < \text{Sn} < \text{Pb}$ as the ns^2 pair becomes inert.
- IV. Sn(II) compounds are reducing while Pb(IV) compounds are oxidising.

V. Pb^{4+} compounds are stronger oxidising agents than Sn^{4+} compounds.

(A) I, II, III and IV only **(B)** I, III, IV and V only
(C) II, III and V only **(D)** all of I–V

Q25. Which option lists exactly the correct statements about the consequences of the inert-pair effect in Group 14?

- I. PbO_2 is a strong oxidising agent.
- II. Lead shows essentially no catenation.
- III. PbI_4 does not exist because I^- reduces Pb(IV) .
- IV. GeO_2 is acidic whereas GeO is amphoteric.

V. PbCl_4 decomposes to $\text{PbCl}_2 + \text{Cl}_2$ on gentle warming.

(A) I, II and III only (B) II, III, IV and V only

(C) I, III and IV only (D) all of I–V

Q26. Which of the following sets contain only amphoteric oxides?

I. Cr_2O_3 , BeO , SnO , SnO_2 .

II. Cr_2O_3 , CrO , SnO , PbO .

III. NO , B_2O_3 , PbO , SnO_2 .

IV. ZnO , Al_2O_3 , PbO , PbO_2 .

V. SiO_2 , CO_2 , GeO_2 , SnO_2 .

(A) I and IV only (B) I, II and IV only

(C) III and IV only (D) I, IV and V only

Q27. Which option lists exactly the correct statements about the formation and acid-base nature of Group 14 oxides?

I. On heating in oxygen the elements give MO and MO_2 ; SiO exists only at high temperature.

II. CO_2 , SiO_2 and GeO_2 are acidic, with acidity $\text{CO}_2 > \text{SiO}_2 > \text{GeO}_2$.

III. SnO_2 and PbO_2 are amphoteric.

IV. CO is a neutral oxide.

V. CO is an acidic oxide, dissolving in water to give carbonic acid directly.

(A) I, II, III and IV only (B) II, III and V only

(C) I, IV and V only (D) all of I–V

Q28. Which option lists exactly the correct statements about reactivity toward water and halogens?

I. C, Si and Ge are unattacked by water; $\text{Sn} + \text{steam} \rightarrow \text{SnO}_2 + \text{H}_2$; Pb is protected by an oxide film.

II. All the tetrahalides except CCl_4 are hydrolysed by water.

III. SiCl_4 hydrolyses because silicon can expand its coordination number using vacant d orbitals.

IV. SnF_4 and PbF_4 are ionic, the other tetrahalides essentially covalent.

V. CCl_4 is hydrolysed by cold water faster than SiCl_4 .

(A) I, II, III and IV only (B) II, III and V only

(C) I, IV and V only (D) all of I–V

Q29. Which option lists exactly the correct statements about higher coordination and complex halide ions?

I. $[\text{SiF}_6]^{2-}$, $[\text{GeCl}_6]^{2-}$ and $[\text{Sn}(\text{OH})_6]^{2-}$ are all six-coordinate.

II. $[\text{SiF}_6]^{2-}$ forms readily but $[\text{SiCl}_6]^{2-}$ does not (F^- small, strong Si–F bonds, less crowding).

III. In these six-coordinate species the central atom is described as sp^3d^2 -hybridised.

IV. Carbon forms neither $[\text{CF}_6]^{2-}$ nor $[\text{CCl}_6]^{2-}$ because it cannot exceed an octet.

V. SiF_4 adds F^- to give $[\text{SiF}_6]^{2-}$, whereas CF_4 shows no such reaction.

(A) I, II and III only (B) II, IV and V only

(C) I, III and IV only (D) all of I–V

Q30. Which option lists exactly the correct statements about the anomalous behaviour of carbon and catenation?

I. Carbon's anomaly stems from small size, high electronegativity, high ionisation enthalpy and absence of d orbitals.

II. Carbon has a maximum covalence of four, whereas the heavier elements can exceed four.

III. Carbon and its heavier congeners show an equal tendency for catenation.

IV. The catenation order is $\text{C} \gg \text{Si} > \text{Ge} \approx \text{Sn}$, and lead shows none.

V. E–E bond enthalpies (kJ mol^{-1}): C–C 348, Si–Si 297, Ge–Ge 260, Sn–Sn 240.

(A) I, II, III and IV only (B) I, II, IV and V only

(C) II, III and V only (D) all of I–V

Q31. Which option lists exactly the correct statements about $\text{p}\pi\text{--d}\pi$ interaction and back bonding in silicon compounds?

I. $\text{N}(\text{SiH}_3)_3$ is planar at nitrogen whereas $\text{N}(\text{CH}_3)_3$ is pyramidal.

II. The Si–O–Si angle in $(\text{SiH}_3)_2\text{O}$ is much larger than the C–O–C angle in $(\text{CH}_3)_2\text{O}$.

III. Both are attributed to $\text{N}\rightarrow\text{Si} / \text{O}\rightarrow\text{Si}$ ($\text{d}\pi$) donation giving partial multiple-bond character.

IV. Trisilylamine is a weaker Lewis base than trimethylamine.

V. Silicon forms stronger $\text{p}\pi\text{--p}\pi$ bonds to oxygen than carbon does.

(A) I, II, III and IV only (B) II, III and V only

(C) I, IV and V only (D) all of I–V

Q32. Which option lists exactly the correct statements about diamond?

I. Each carbon is sp^3 -hybridised and tetrahedrally bonded to four others; $\text{C--C} = 154 \text{ pm}$.

II. Diamond is a rigid three-dimensional covalent network, extremely hard, with a very high melting point.

III. Diamond is a layered solid held together by van der Waals forces.

IV. Diamond does not conduct electricity.

V. Diamond is used as an abrasive, for cutting tools and in dies.

(A) I, II, IV and V only (B) II, III and V only

(C) I, II, III and IV only (D) all of I–V

Q33. Which option lists exactly the correct statements about graphite?

I. Graphite is layered; the layers are 340 pm apart and held by weak van der Waals forces.

II. Within a layer the C–C bond length is 141.5 pm and each carbon is sp^2 .

III. Graphite has a three-dimensional covalent network like diamond and is used as an abrasive.

IV. The delocalised fourth electron makes graphite an electrical conductor along the layers.

V. Graphite is soft and slippery and is used as a lubricant and for electrodes.

(A) I, II, III and V only (B) I, II, IV and V only

(C) II, III and IV only (D) all of I–V

Q34. Which option lists exactly the correct statements about fullerenes?

I. Fullerenes are made by heating graphite in an electric arc in He or Ar.

II. They are the only pure form of carbon, with no dangling bonds.

III. C₆₀ has twenty six-membered and twelve five-membered rings; every carbon is sp².

IV. In C₆₀ every five-membered ring is fused only to other five-membered rings.

V. The two C–C bond lengths in C₆₀ are 143.5 pm and 138.3 pm.

(A) I, II, III and IV only (B) II, III and V only

(C) I, II, III and V only (D) all of I–V

Q35. Which option lists exactly the correct statements about fullerene bonding?

I. Fullerenes are non-planar cage molecules.

II. The 6:6 ring-fusion bonds are shorter (more double-bond character) than the 6:5 bonds.

III. Fullerenes are less aromatic than benzene.

IV. Every C_n fullerene (n even, ≥ 20) has exactly 12 pentagons and (n – 20)/2 hexagons.

V. Fullerenes possess dangling bonds at the pentagon vertices.

(A) I, II, III and IV only (B) II, III and IV only

(C) I, III, IV and V only (D) all of I–V

Q36. Which option lists exactly the correct statements about carbon-allotrope energetics and structural data?

I. ΔfH°: graphite 0, diamond +1.9, C₆₀ +38.1 kJ mol^{–1}.

II. C₆₀ has 20 pentagons and 12 hexagons.

III. C–C distances: diamond 154 pm, graphite in-plane 141.5 pm, graphite interlayer 340 pm.

IV. For diamond → graphite at 298 K, ΔH < 0 and ΔS > 0.

V. C₆₀ has 32 rings in all and 90 C–C edges.

(A) I, II, III and V only (B) II, III and IV only

(C) I, III, IV and V only (D) all of I–V

Q37. Which option lists exactly the correct statements about impure forms of carbon and carbon materials?

I. Carbon black is made by burning hydrocarbons in a limited air supply.

II. Coke and charcoal are obtained by heating coal and wood in the absence of air.

III. Activated charcoal adsorbs poisonous gases and purifies water because it is highly porous.

IV. An ordinary “lead” pencil contains about 50% metallic lead.

V. Graphite-fibre composites are strong and light; graphite crucibles resist dilute acids and alkalis.

(A) I, II, III and IV only (B) II, III and V only

(C) I, II, III and V only (D) all of I–V

Q38. Which option lists exactly the correct statements about carbon monoxide preparation and industrial gases?

I. CO is formed by incomplete combustion of carbon; pure CO from HCOOH + conc. H₂SO₄ at 373 K.

II. Steam over hot coke gives water gas (CO + H₂); air over hot coke gives producer gas (CO + N₂).

III. Water gas is a mixture of CO₂ and H₂.

IV. Producer gas is a less efficient fuel than water gas.

V. The CO of water gas can be shifted with steam to CO₂ + H₂.

(A) I, II, III and IV only (B) II, III and V only

(C) I, II, IV and V only (D) all of I–V

Q39. Which option lists exactly the correct statements about carbon monoxide bonding, reducing action and toxicity?

I. CO is isoelectronic with N₂, CN[–] and NO⁺; it has one σ and two π components.

II. CO is a C-donor in metal carbonyls (e.g. Ni(CO)₄), with back-donation into its π* orbitals.

III. CO reduces Fe₂O₃ and ZnO on heating and can be estimated by oxidation with I₂O₅.

IV. Carboxyhaemoglobin is about 300 times more stable than the oxygen–haemoglobin complex.

V. CO is an acidic oxide, forming H₂CO₃ with water at room temperature.

(A) I, II, III and IV only (B) II, III and V only

(C) I, III and IV only (D) all of I–V

Q40. Which option lists exactly the correct statements about carbon dioxide preparation and acid-base chemistry?

I. CO₂ is made in the laboratory from CaCO₃ + dilute HCl and commercially by heating limestone.

II. Dissolved CO₂ gives weak dibasic H₂CO₃, ionising stepwise via HCO₃[–] and CO₃^{2–}.

III. The H₂CO₃/HCO₃[–] system buffers blood near pH 7.26–7.42.

IV. With alkali CO₂ gives carbonate, or with 1 : 1 CO₂ : NaOH, hydrogencarbonate.

V. Carbonic acid is a strong dibasic acid.

(A) I, II, III and IV only (B) II, III and V only

(C) I, III, IV and V only (D) all of I–V

Q41. Which option lists exactly the correct statements about carbon dioxide structure, uses and environmental chemistry?

I. Carbon in CO₂ is sp-hybridised; the molecule is linear (C–O ≈ 115 pm), zero dipole, isostructural with N₃[–] and N₂O.

II. Dry ice is a molecular solid used as a refrigerant; CO₂ is used in fire extinguishers and to make urea.

III. CO₂ is bent because carbon carries a lone pair.

IV. CO₂ is removed from the atmosphere by photosynthesis; rising CO₂ intensifies the greenhouse effect.

V. Solid CO₂ is molecular (not a network) because carbon forms effective pπ–pπ bonds to oxygen.

(A) I, II, III and IV only (B) I, II, IV and V only

(C) II, III and V only (D) all of I–V

Q42. Which option lists exactly the correct reactions involving carbon and its oxides?

I. $2C + O_2(\text{limited}) \rightarrow 2CO$; $C + O_2(\text{excess}) \rightarrow CO_2$.

II. $HCOOH \xrightarrow{\text{conc. } H_2SO_4, 373\text{ K}} CO + H_2O$.

III. $H_2C_2O_4 \xrightarrow{\text{conc. } H_2SO_4, \Delta} CO + CO_2 + H_2O$.

IV. $C + H_2O(g) \xrightarrow{\text{red heat}} CO + H_2$ (water gas).

V. $C + H_2O(g) \xrightarrow{\text{red heat}} CO_2 + H_2$.

(A) I, II, III and IV only (B) II, III and V only

(C) I, III, IV and V only (D) all of I–V

Q43. Which option lists exactly the correct statements about carbon suboxide and phosgene?

I. Carbon suboxide is C₃O₂ (O=C=C=O), from malonic acid, linear with sp carbons.

II. C₃O₂ contains 4 σ and 4 π bonds (8 in total).

III. $CO + Cl_2 \rightarrow COCl_2$; $COCl_2 + 4NH_3 \rightarrow CO(NH_2)_2 + 2NH_4Cl$.

IV. Heating oxalic acid with conc. H₂SO₄ gives CO + CO₂ + H₂O.

V. Dehydration of formic acid with conc. H₂SO₄ gives CO₂.

(A) I, II, III and IV only (B) II, III and V only

(C) I, III and IV only (D) all of I–V

Q44. Which option lists exactly the correct statements about carbides?

I. CaC₂ (acetylide) → C₂H₂; Al₄C₃ and Be₂C (methanides) → CH₄; Mg₂C₃ → C₃H₄ on hydrolysis.

II. SiC (carborundum) is a covalent-network abrasive; B₄C is an armour ceramic; WC is a hard interstitial carbide.

III. WC is a discrete molecular carbide that dissolves readily in water.

IV. $CaC_2 + N_2 \xrightarrow{\Delta} CaCN_2 + C$.

V. Beryllium carbide gives methane on hydrolysis, confirming its methanide nature.

(A) I, II, III and IV only (B) II, III and V only

(C) I, II, IV and V only (D) all of I–V

Q45. For which of the following metals is the carbon-based reduction method NOT used for extraction?

I. Sn from SnO₂.

II. Fe from Fe₂O₃.

III. Al from Al₂O₃.

IV. Mg from MgCO₃ · CaCO₃ (dolomite).

V. Zn from ZnO.

(A) III and IV only (B) I and II only (C) III, IV and V only (D) I, III and IV only

Q46. Which option lists exactly the correct statements about the forms, structure and reactions of silica?

I. Quartz, cristobalite and tridymite are crystalline forms; kieselguhr is amorphous.

II. SiO₂ is a covalent 3-D network: each Si bonds four O, each O bridges two Si.

III. In SiO₂ silicon forms discrete Si=O double bonds, as carbon does in CO₂.

IV. Silica is attacked by HF (→ SiF₄, then H₂SiF₆) and by hot NaOH (→ Na₂SiO₃) but not by other mineral acids.

V. Quartz is piezoelectric; silica gel is a drying agent and chromatographic support.

(A) I, II, III and IV only (B) II, III and V only

(C) I, II, IV and V only (D) all of I–V

Q47. Which of the following statements about silicone precursor functionality and polymer architecture are correct?

Silicone precursor functionality and polymer architecture can be controlled by the nature of the chlorosilane used. Consider the following statements regarding the preparation and structural control of silicones:

I. In the direct process, methyl chloride reacts with elemental silicon in the presence of Cu catalyst at about 573 K to produce methylchlorosilanes.

II. Hydrolysis of R₂SiCl₂ gives a difunctional silanol species which, on condensation, predominantly favours linear or cyclic siloxane structures.

III. RSiCl₃ is trifunctional with respect to hydrolysable Si–Cl bonds and can therefore introduce branching and three-dimensional cross-linking into a silicone structure.

IV. R₃SiCl can undergo hydrolysis and condensation, but after incorporation its only reactive functionality is consumed; hence, it can act as an end-blocking or chain-terminating unit.

V. R₄Si cannot independently form a silicone network through the usual hydrolysis–condensation route because it contains no hydrolysable Si–Cl bond.

(A) I, II and III only (B) II, III, IV and V only

(C) I, II, III and IV only (D) I, II, III, IV and V

Q48. Which option lists exactly the correct statements about the properties and applications of silicones?

I. Silicones are hydrophobic because non-polar organic groups shield the Si–O backbone.

II. They have high thermal stability and resist oxidation and most organic reagents.

III. Silicones are strongly hydrophilic, water-soluble ionic polymers of low dielectric strength.

IV. They are used as sealants, greases/lubricants, electrical insulators, water-proofing and antifoaming agents.

V. Being biocompatible, they are used in surgical and cosmetic implants.

(A) I, II, III and IV only (B) I, II, IV and V only

(C) II, III and V only (D) all of I–V

Q49. Which option lists exactly the correct statements about the basic unit and polymerisation of silicates?

I. The basic unit is the $[\text{SiO}_4]^{4-}$ tetrahedron.

II. Tetrahedra may be discrete or share 1–4 corner oxygens, giving chains, rings, sheets or frameworks.

III. The negative charge is balanced by metal cations; feldspar, mica, asbestos and zeolites are natural silicates; glass and cement are man-made.

IV. Silicates are restricted to isolated $[\text{SiO}_4]^{4-}$ ions and cannot polymerise.

V. Sharing all four corners gives a neutral framework, SiO_2 .

(A) I, II, III and V only (B) II, III and IV only

(C) I, III, IV and V only (D) all of I–V

Q50. Which option lists exactly the correct statements about silicate types and their idealised formulae?

I. Orthosilicate $[\text{SiO}_4]^{4-}$; pyrosilicate $[\text{Si}_2\text{O}_7]^{6-}$ (one shared corner).

II. Cyclic silicates $[\text{Si}_3\text{O}_9]^{6-}$ and $[\text{Si}_6\text{O}_{18}]^{12-}$ have two shared corners per tetrahedron.

III. Single chain $(\text{SiO}_3)_n^{2n-}$; double chain $(\text{Si}_4\text{O}_{11})_n^{6n-}$ (average 2.5 shared corners).

IV. Sheet silicate $(\text{Si}_2\text{O}_5)_n^{2n-}$ (three shared corners); framework SiO_2 (four shared corners).

V. A single chain and a sheet silicate have the same O : Si ratio.

(A) I, II, III and V only (B) II, III and IV only

(C) I, II, III and IV only (D) all of I–V

Q51. Which option lists exactly the correct periodic and bonding trends in Group 14?

I. Catenation: $\text{C} \gg \text{Si} > \text{Ge} \approx \text{Sn} > \text{Pb}$.

II. First ionisation enthalpy: $\text{C} > \text{Si} > \text{Ge} > \text{Pb} > \text{Sn}$.

III. Acidic character of the dioxides: $\text{CO}_2 > \text{SiO}_2 > \text{GeO}_2$.

IV. Reducing power of the hydrides: $\text{CH}_4 < \text{SiH}_4 < \text{GeH}_4 < \text{SnH}_4$; MX_2 stability increases down the group.

V. CCl_4 boils above SiCl_4 , and graphite is harder than diamond.

(A) I, II, III and V only (B) II, III and IV only

(C) I, II, III and IV only (D) all of I–V

Q52. Which option lists exactly the correct statements about zeolites and aluminosilicates?

I. Replacing some Si by Al gives a net negative framework, balanced by Na^+ , K^+ or Ca^{2+} .

II. Zeolites are shape-selective catalysts for cracking and isomerisation; ZSM-5 converts alcohols to gasoline.

III. Hydrated zeolites soften hard water by exchanging framework Na^+ for $\text{Ca}^{2+}/\text{Mg}^{2+}$.

IV. Al-for-Si substitution makes the framework positive and softening is by anion exchange.

V. $\text{Na}_2\text{SiO}_3 + 2\text{HCl} \rightarrow \text{hydrated silica / silicic-acid gel} + 2\text{NaCl}$.

(A) I, II, III and V only (B) II, III and IV only (C) I, III, IV and V only (D) all of I–V

Q53. Which option lists exactly the correct statements about the descriptive chemistry of tin and lead?

I. “Tin pest” is the crumbling of white β -tin to grey α -tin below about 13 °C.

II. Butter of tin is $\text{SnCl}_4 \cdot 5\text{H}_2\text{O}$; litharge PbO ; red lead Pb_3O_4 ; white lead $2\text{PbCO}_3 \cdot \text{Pb}(\text{OH})_2$.

III. Galena (PbS), anglesite (PbSO_4) and cerussite (PbCO_3) are lead ores; chrome yellow is PbCrO_4 .

IV. Solder is an alloy of tin and lead.

V. On heating with conc. H_2SO_4 , tin gives H_2S as the sulfur product.

(A) I, II, III and V only (B) II, III and IV only

(C) I, II, III and IV only (D) all of I–V

Q54. Which option lists exactly the correct statements about lead oxides and Sn(II)/Pb(IV) redox chemistry?

I. $\text{Pb}_3\text{O}_4 \equiv 2\text{PbO} \cdot \text{PbO}_2$.

II. $\text{Pb}_3\text{O}_4 + 8\text{HCl}(\text{conc.}) \rightarrow 3\text{PbCl}_2 + \text{Cl}_2 + 4\text{H}_2\text{O}$.

III. $2\text{Pb}(\text{NO}_3)_2 \xrightarrow{\Delta} 2\text{PbO} + 4\text{NO}_2 + \text{O}_2$.

IV. $\text{SnCl}_2 + 2\text{FeCl}_3 \rightarrow \text{SnCl}_4 + 2\text{FeCl}_2$; SnCl_2 reduces HgCl_2 to Hg_2Cl_2 then to Hg .

V. PbO_2 acts as a reducing agent toward concentrated HCl .

(A) I, II, III and V only (B) II, III and IV only

(C) I, II, III and IV only (D) all of I–V

SECTION C — SINGLE CORRECT QUESTIONS (Q55–Q70)

Q55. The tendency for catenation is maximum in:

(A) Pb (B) Ge (C) Si (D) C

Q56. White metallic tin, kept in extreme cold, crumbles to a grey powder (the “tin disease” that ruined the buttons of Napoleon’s soldiers). This transformation is due to:

(A) interaction with nitrogen of the air at very low temperature
(B) a change in the crystalline structure of tin
(C) a change in the partial pressure of oxygen in the air
(D) interaction with water vapour in the humid air

Q57. Which species has the same bond order as CO ?

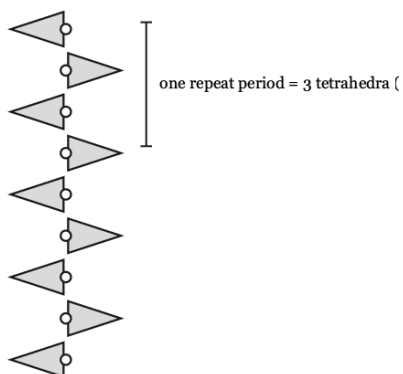
(A) O_2^- (B) NO (C) NO^+ (D) O_2

Q58. SiCl_4 on hydrolysis gives X and HCl ; on strong heating ($\approx 1000\text{ K}$) X loses water to give Y. X and Y are, respectively:

(A) H_2SiCl_6 and SiO_2 (B) H_4SiO_4 and Si

(C) H_4SiO_4 and SiO_2 (D) SiO_2 and Si

Q59. Single-chain (pyroxene / metasilicate) silicates are classified by their repeat period nT — the chain repeats after n SiO_4 tetrahedra. Examples are CuGeO_3 (1T), Na_2SiO_3 ("long-2T"), BaSi_2O_6 ("short-2T"), $\text{Ca}_3\text{Si}_3\text{O}_9$ = wollastonite (3T) and $\text{Ba}_2\text{Si}_4\text{O}_8(\text{OH})_4 \cdot 4\text{H}_2\text{O}$ ("long-4T"). Every one of these chains has the anion formula:



single-chain (pyroxene / metasilicate) silicate: each SiO_4 tetrahedron shares 2 corners; anion $(\text{SiO}_3)_n$, charge $2n$ minus

single-chain silicate: $3T$ repeat

(A) $(\text{SiO}_4)_n^{4n-}$ (each tetrahedron shares 0 corners) (B) $(\text{SiO}_3)_n^{2n-}$ (each tetrahedron shares 2 corners) (C) $(\text{Si}_2\text{O}_5)_n^{2n-}$ (each tetrahedron shares 3 corners) (D) $(\text{Si}_4\text{O}_{11})_n^{6n-}$ (each tetrahedron shares 2.5 corners)

Q60. Boron cannot form BF_6^{3-} and carbon cannot form CCl_6^{2-} , although Al forms AlF_6^{3-} and Si forms SiF_6^{2-} . The common reason is:

(A) the absence of valence-shell d orbitals in B and C (B) the low electronegativity of B and C (C) the large size of B and C (D) the high ionisation enthalpy of F and Cl

Q61. "Dry ice" is:

(A) solid NH_3 (B) solid SO_2 (C) solid CO_2 (D) solid N_2

Q62. Among C, Si, Ge, Sn and Pb, the element with the greatest tendency to form $\text{p}\pi\text{--}\text{p}\pi$ multiple bonds is:

(A) Si (B) Ge (C) Sn (D) C

Q63. The Group 14 element that liberates H_2 on reaction with steam is:

(A) Si (B) Ge (C) Sn (D) Pb

Q64. Which option gives the correct True (T) / False (F) sequence for the following statements?

A. Heavier Group 14 elements do not form $\text{p}\pi\text{--}\text{p}\pi$ bonds because their atomic orbitals are not too large. B. The order of catenation is $\text{C} > \text{Si} > \text{Ge} \approx \text{Sn}$. C. Diamond is a covalent (giant) molecule and therefore has a high melting point. D. In fullerenes a five-membered ring fuses only with six-membered rings.

(A) T F T T (B) T T F T (C) F T F T (D) F T T T

Q65. Which comparison of hydrolysis is correct?

(A) both CCl_4 and SiCl_4 are readily hydrolysed (B) CCl_4 hydrolyses readily but SiCl_4 does not (C) CCl_4 resists hydrolysis but SiCl_4 hydrolyses readily (D) neither is hydrolysed

Q66. Carbon suboxide, obtained by dehydrating malonic acid with P_4O_{10} , is:

(A) C_2O (B) C_2O_3 (C) C_3O_4 (D) C_3O_2

Q67. When SiMe_3Cl is co-hydrolysed with an excess of SiMe_2Cl_2 and the product is condensed, the silicone obtained is:

(A) a highly cross-linked three-dimensional network

(B) a short linear siloxane chain end-capped by $\text{Me}_3\text{Si-}$ groups

(C) a purely cyclic tetramer

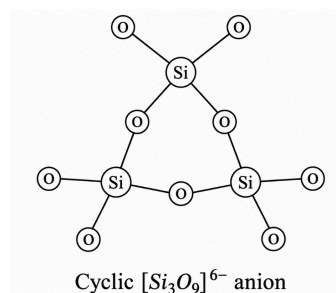
(D) crystalline SiO_2

Q68. The silicate anion in kaolinite (china clay) is a sheet in which each SiO_4 tetrahedron shares three corners. Kaolinite is:

(A) Al_2SiO_4 (B) $\text{AlSiO}_3(\text{OH})$

(C) $\text{Al}_2\text{Si}_4\text{O}_{10}(\text{OH})_2$ (D) $\text{Al}_2\text{Si}_2\text{O}_5(\text{OH})_4$

Q69. In the silicate anion shown, each SiO_4 tetrahedron shares two of its corner oxygen atoms with neighbours. The anion is:



(A) an open, branched chain of three tetrahedra

(B) a discrete double tetrahedron, $\text{Si}_2\text{O}_7^{6-}$

(C) a closed three-membered ring, $\text{Si}_3\text{O}_9^{6-}$

(D) a fragment of an infinite sheet

Q70. Litharge and red lead are, respectively:

(A) PbO_2 and PbO (B) Pb_3O_4 and PbO_2

(C) PbO and PbO_2 (D) PbO and Pb_3O_4

SECTION D — INTEGER TYPE (Q71–Q82)

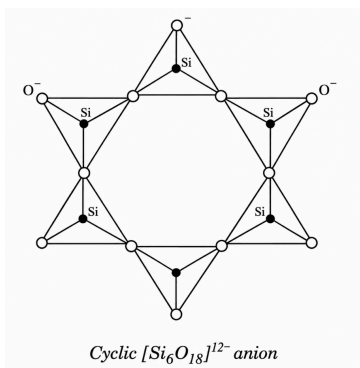
Answer is a single non-negative integer.

Q71. In C_{70} fullerene, x = number of five-membered rings and y = number of six-membered rings. Find $(y - x)$.

Q72. (i) Density of graphite = $x \text{ g cm}^{-3}$, of diamond = $y \text{ g cm}^{-3}$; A = nearest integer of $(x + y) \cdot 10$. (ii) P = dative bonds in $(\text{AlCl}_3)_2$; Q = dative bonds in $[\text{Al}(\text{H}_2\text{O})_6]\text{Cl}_3$. Find $A + P + Q$.

Q73. How many of the Group 13 and Group 14 elements have a higher IE_1 than Pb?

Q74. The cyclic silicate anion shown contains x silicon atoms and y oxygen atoms and carries an overall charge of $z-$. Calculate $|x + y - z|$.



Q75. (a) Among C_{60} , graphite, CO_2 , C_3O_2 , CO , CO_3^{2-} and diamond, the number of species with sp -hybridised carbon is x .
 (b) Be_2C_p and Al_4C_q are methanides. Find $x + p + q$.

Q76. How many of the following are correctly matched?

- (i) $SiF_4 \Rightarrow$ gaseous, tetrahedral molecule (ii) $Si \Rightarrow$ semiconductor (iii) $Cs_3C_{60} \Rightarrow$ superconducting near 40 K (iv) $SnO \Rightarrow$ amphoteric oxide (v) $[SiO_4]^{4-} \Rightarrow$ its calcium salt is a component of cement (vi) $PbO \Rightarrow$ acidic oxide

Q77. For CO , SiO_2 , B_2O_3 , PbO_2 , SnO_2 , PbO , SnO , CO_2 , Al_2O_3 : $x =$ neutral, $y =$ basic, $z =$ acidic, $t =$ amphoteric oxides. Find $|2x + y - z - t|$.

Q78. The number of $3c-2e^-$ bonds present in $[Al(BH_4)_4]^-$ (each BH_4 bidentate to Al) is:

Q79. The number of boron atoms in colemanite, $Ca_2B_6O_{11} \cdot 5H_2O$, that have **no** back bonding (tetrahedral BO_4 boron) is:

Q80. Among $Sc_2[Si_2O_7]$, $Be_3Al_2[Si_6O_{18}]$, $LiAlSi_2O_6$, $Na_2Fe_3Fe_2Si_8O_{22}(OH)_2$ (riebeckite) and $Mg_3Si_4O_{10}(OH)_2$ (talca), the number of silicates in which every SiO_4 tetrahedron shares at least two oxygen atoms is:

Q81. How many of the following are paramagnetic: Fe_2O_3 , MnO , La_2O_3 , ZnO , $AgCl$, SO_3 , PCl_5 , CO , NO , PmO , Eu_2O_3 , U_2O_3 , MnO_2 ?

Q82. The sum of the numbers of pentagonal rings in C_{40} , C_{60} and C_{80} is (give the repeated digit-sum single digit):

SECTION E — MATRIX MATCH (Q83–Q92)

Q83. Match each fuel gas in Column-I with all of its major constituent(s) in Column-II.

Column-I	Column-II
I) Water gas	P) CO
II) Producer gas	Q) N_2
III) Coal gas	R) H_2
IV) Natural gas	S) CH_4
	T) CO_2

(A) I–(P,Q); II–(P,R); III–(Q,R,S); IV–(S,T)

(B) I–(R); II–(P,Q); III–(P,R); IV–(S)

(C) I–(P,R); II–(P,Q); III–(P,R,S); IV–(S)

(D) I–(P,R,S); II–(P,Q); III–(R,S); IV–(P,S)

Q84. Match each silicate class in Column-I with all applicable structural features in Column-II.

Column-I	Column-II
I) Orthosilicate	P) built from SiO_4 tetrahedra
II) Pyrosilicate	Q) no corner O shared per

Column-I	Column-II
III) Pyroxene (single chain)	tetrahedron
IV) Mica (sheet)	R) one corner O shared per tetrahedron
	S) two corner O shared per tetrahedron
	T) three corner O shared per tetrahedron
	U) polymeric/infinite anion

(A) I–(P); II–(P,S); III–(P,R,U); IV–(P,T)

(B) I–(P,Q); II–(P,R); III–(P,S,U); IV–(P,T,U)

(C) I–(P,Q,U); II–(P,R,U); III–(P,S); IV–(P,T)

(D) I–(P,T); II–(P,Q); III–(P,S,U); IV–(P,R,U)

Q85. Match each species in Column-I with all applicable central-atom hybridisation / geometry descriptors in Column-II.

Column-I	Column-II
I) CO_2	P) sp
II) CO_3^{2-}	Q) linear
III) $[SiF_6]^{2-}$	R) sp^2
IV) $SiCl_4$	S) trigonal planar
	T) sp^3
	U) tetrahedral
	V) sp^3d^2
	W) six-coordinate

(A) I–(P,S); II–(R,Q); III–(V,U); IV–(T,W)

(B) I–(P,Q); II–(R,S); III–(T,U); IV–(V,W)

(C) I–(P,Q); II–(T,U); III–(V,W); IV–(R,S)

(D) I–(P,Q); II–(R,S); III–(V,W); IV–(T,U)

Q86. Match each oxide in Column-I with all applicable descriptions in Column-II.

Column-I	Column-II
I) CO	P) neutral oxide
II) CO_2	Q) acidic oxide
III) SnO	R) amphoteric oxide
IV) PbO_2	S) strong oxidising agent
	T) monoxide

(A) I–(P,T); II–(Q); III–(R,T); IV–(R,S)

(B) I–(Q,T); II–(P); III–(R); IV–(S,T)

(C) I–(P); II–(Q,T); III–(R,S); IV–(R)

(D) I–(R,T); II–(Q); III–(P,T); IV–(R,S)

Q87. Match each Group 14 halide / complex in Column-I with all applicable descriptions in Column-II.

Column-I	Column-II
I) CCl_4	P) tetrahedral molecular halide
II) $SiCl_4$	Q) resists hydrolysis under ordinary conditions
III) $[SiF_6]^{2-}$	R) hydrolyses readily
IV) $[GeCl_6]^{2-}$	S) six-coordinate species
	T) central-atom coordination number > 4

(A) I–(P,R); II–(P,Q); III–(S); IV–(T)

(B) I–(Q); II–(P,R,T); III–(S,T); IV–(S)

(C) I–(P,Q); II–(P,R); III–(S,T); IV–(S,T)

(D) I–(P,Q,T); II–(R); III–(S); IV–(P,S,T)

Q88. Match each carbon form in Column-I with all applicable structural / use descriptors in Column-II.

Column-I	Column-II
I) Diamond	P) extended three-dimensional covalent network
II) Graphite	Q) sp^2 carbon
III) C_{60}	R) delocalised electrons responsible for conduction
IV) Carbon black	S) cage molecule
	T) pigment / tyre filler
	U) sp^3 carbon

(A) I–(P,Q); II–(R,U); III–(Q,S); IV–(T)

(B) I–(P,U); II–(Q,R); III–(Q,S); IV–(T)

(C) I–(U); II–(P,Q,R); III–(S); IV–(Q,T)

(D) I–(P,U); II–(Q); III–(Q,R,S); IV–(T)

Q89. Match each species / mixture in Column-I with all applicable descriptions in Column-II.

Column-I	Column-II
I) CO	P) neutral oxide
II) CO_2	Q) acidic oxide
III) H_2CO_3	R) reducing component
IV) Water gas	S) contains H_2 as a major constituent
	T) weak dibasic acid

(A) I–(P); II–(Q,R); III–(T); IV–(S)

(B) I–(Q,R); II–(P); III–(T,S); IV–(R)

(C) I–(P,R,S); II–(Q); III–(T); IV–(R)

(D) I–(P,R); II–(Q); III–(T); IV–(R,S)

Q90. Match each organosilicon precursor in Column-I with all applicable roles / features in Column-II.

Column-I	Column-II
I) R_2SiCl_2	P) contains hydrolysable Si–Cl bond
II) $RSiCl_3$	Q) mainly gives linear / cyclic siloxanes
III) R_3SiCl	R) can introduce cross-linking
IV) R_4Si	S) chain-terminating / end-blocking unit
	T) no hydrolysable Si–Cl bond

(A) I–(P,Q); II–(P,R); III–(P,S); IV–(T)

(B) I–(P,R); II–(P,Q); III–(S); IV–(P,T)

(C) I–(Q); II–(R); III–(P,S,T); IV–(P)

(D) I–(P,Q,S); II–(R); III–(P); IV–(T)

Q91. Match each tin / lead species in Column-I with all applicable descriptions in Column-II.

Column-I	Column-II
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Column-I	Column-II
I) $SnCl_2$	P) reducing agent
II) PbO_2	Q) oxidising agent
III) Pb_3O_4	R) amphoteric oxide
IV) PbO	S) mixed Pb(II)/Pb(IV) oxide
	T) litharge
	U) red lead

(A) I–(Q); II–(P,R); III–(S,T); IV–(R,U)

(B) I–(P,R); II–(Q); III–(S,U); IV–(T)

(C) I–(P); II–(Q,R); III–(S,U); IV–(R,T)

(D) I–(P); II–(R); III–(Q,S,U); IV–(R,T)

Q92. Match each silicate anion in Column-I with all applicable structural descriptions in Column-II.

Column-I	Column-II
I) $[SiO_4]^{4-}$	P) isolated tetrahedron
II) $[Si_2O_7]^{6-}$	Q) one shared corner per tetrahedron
III) $[Si_3O_9]^{6-}$	R) two shared corners per tetrahedron
IV) $(Si_2O_5)_n^{2n-}$	S) three shared corners per tetrahedron
	T) cyclic anion
	U) sheet silicate

(A) I–(P,Q); II–(R); III–(T); IV–(S,U)

(B) I–(P); II–(Q); III–(R,T); IV–(S,U)

(C) I–(P); II–(Q,T); III–(R); IV–(S,U)

(D) I–(P,U); II–(Q); III–(R,T); IV–(S)

SECTION F — COMPREHENSION (Q93–Q96)

Paragraph for Q93–Q94

A bright red solid **A** dissolves in warm dilute HNO_3 to give a solution of a lead(II) salt and leaves a dark brown solid **B**. The same solid **A**, treated with concentrated HCl , liberates Cl_2 gas and leaves a white residue **C**.

Q93. The solids A and B are, respectively:

(A) Pb_3O_4 and PbO_2 (B) PbO_2 and Pb_3O_4 (C) Pb_3O_4 and PbO (D) Pb_2O_3 and PbO_2

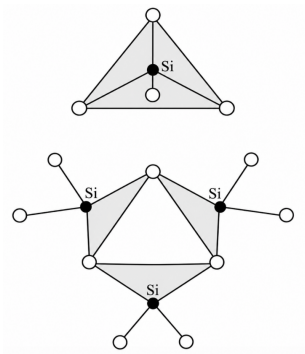
Q94. The white residue C is:

(A) $PbCl_2$ (B) $PbCl_4$ (C) PbO (D) $Pb(NO_3)_2$

Paragraph for Q95–Q96

“Silica” covers a group of minerals of general formula SiO_2 , the commonest being quartz, a framework silicate built from corner-sharing SiO_4 tetrahedra. Silicate anions are classified by the number of corner oxygens each tetrahedron shares.

Q95. The two silicate-anion diagrams below show (i) a single isolated tetrahedron and (ii) three tetrahedra sharing corners in a closed ring. Their formulae are, respectively:



(A) SiO_3^{2-} and $\text{Si}_3\text{O}_7^{2-}$ (B) SiO_4^{4-} and $\text{Si}_3\text{O}_{10}^{8-}$ (C) SiO_4^{4-} and $\text{Si}_3\text{O}_9^{6-}$ (D) SiO_3^{2-} and $\text{Si}_3\text{O}_9^{6-}$

Q96. A mineral whose silicate anion is a chain of three SiO_4 tetrahedra, and which also contains Ca^{2+} , Cu^{2+} and water in the ratio 2 : 2 : 2, is:

(A) $\text{CaCuSi}_3\text{O}_{10} \cdot \text{H}_2\text{O}$ (B) $\text{CaCuSi}_3\text{O}_{10} \cdot 2\text{H}_2\text{O}$ (C) $\text{Ca}_2\text{Cu}_2\text{Si}_3\text{O}_{10} \cdot 2\text{H}_2\text{O}$ (D) $\text{Ca}_3\text{Cu}_3\text{Si}_3\text{O}_{10} \cdot 3\text{H}_2\text{O}$

Section A — Number of Correct Statements

Answer Key

Question Correct Question Correct

Q1	10	Q11	11
Q2	10	Q12	10
Q3	10	Q13	10
Q4	10	Q14	11
Q5	10	Q15	11
Q6	10	Q16	11
Q7	10	Q17	10
Q8	10	Q18	10
Q9	11	Q19	10
Q10	12	Q20	10

Section B — Q21–Q54

Question Answer Question Answer

Q21	C	Q38	C
Q22	A	Q39	A
Q23	B	Q40	A
Q24	B	Q41	B
Q25	D	Q42	A
Q26	A	Q43	A
Q27	A	Q44	C
Q28	A	Q45	A
Q29	D	Q46	C
Q30	B	Q47	D
Q31	A	Q48	B
Q32	A	Q49	A

Question Answer Question Answer

Q33	B	Q50	C
Q34	C	Q51	C
Q35	A	Q52	A
Q36	C	Q53	C
Q37	C	Q54	C

Section C — Q55–Q70

Question Answer Question Answer

Q55	D	Q63	C
Q56	B	Q64	D
Q57	C	Q65	C
Q58	C	Q66	D
Q59	B	Q67	B
Q60	A	Q68	D
Q61	C	Q69	C
Q62	D	Q70	D

Section D — Integer Type, Q71–Q82

Question Answer Question Answer

Q71	13	Q77	6
Q72	65	Q78	8
Q73	4	Q79	4
Q74	12	Q80	4
Q75	7	Q81	7
Q76	5	Q82	9

Section E — Matrix Match, Q83–Q92

Question Answer Question Answer

Q83	C	Q88	B
Q84	B	Q89	D
Q85	D	Q90	A
Q86	A	Q91	C
Q87	C	Q92	B

Section F — Comprehension, Q93–Q96

Question Answer

Q93	A
Q94	A
Q95	C
Q96	C